LOURDES MATHA COLLEGE OF SCIENCE & TECHNOLOGY

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22/30

ANSWER BOOK

e Athullyamol S R Class No 14 Branch MCA

ct Design & Analysis of Algorithms. Semester S3

Mark Tabulation

	Q.1	Q.2	Q.3	Q.4	Q.5	Q.6	Q.7	Q.8	Q.9	Q.10	Q.11	Q.12	Q.13	Q.14	
	3	–	3	3	1	3	6	3							
al															
Grand Total													22		

Part - B.

The knapsack algorithm is a greedy method algorithm.

The objective of knapsack algorithm is to fill knapsack to obtain an maximum profit.

Given,

n :- Number of items = 3

M :- knapsack capacity = 5

Pi : Profit of i item

wi :. Weight of i item

Xi :- fraction part of i th item.

Item	P1	P2	P3
Pi	500	900	700
wi	1	3	7
Pi/wi	500	300	100

Step 1 :- we want to arrange items in

based on Pi/wi :- Profit - weig

Item	P1	P2	P3
Pi	500	900	700
wi	1	3	7



Step 2:- Initialize the fraction set x_i as 0.

$x_i = \{0, 0, 0\}$

Step 3:- Take first item P_1 and check the weight of

P_1 is less than the capacity of knapsack.

ie, $1 \leq 5$, therefore the fraction set

upgrade to

$x_i = \{1, 0, 0\}$

Step 4:- Take next item P_2 and check the weigh

is less than the capacity of knapsack.

ie, $1 + 3 \leq 5$

$4 \leq 5$

Here in the capacity of knapsack

occupied.

So the fraction set

$x_i = \{1, 1, 0\}$

7,

ie $1+3+7 \leq 5$

$11 \leq 5$

11 is greater than the capacity of

knapsack.

It exceeds from the capacity of knapsack which is against the constraint, so we take part of item P_3. So the fraction set become

$$x_i = \{1, 1, 1/7\}$$

So the profit of 3 item are

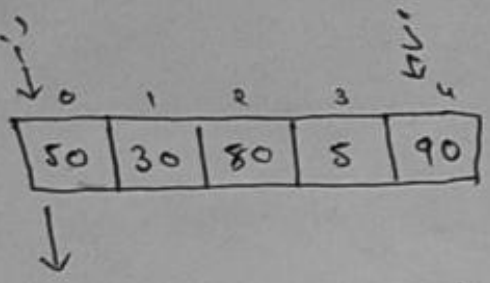
$$1 \times 500 + 1 \times 900 \times \frac{1}{7} \times 700$$

$$= 500 + 900 + 100$$

$$= 1500$$

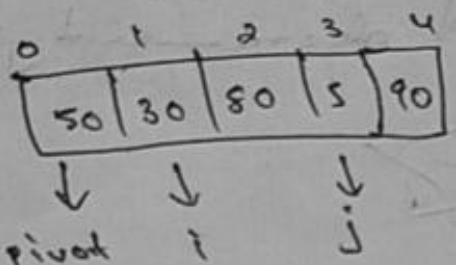
The maximum profit of 3 item = 15

iven

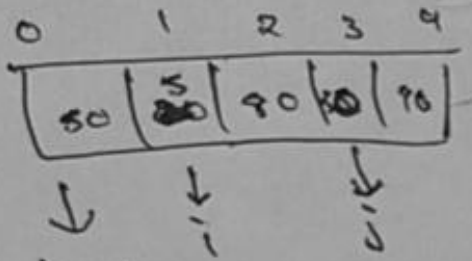


Pivot = 50 i = 80 j = 4.

Increment the index i to 1 and decrement the



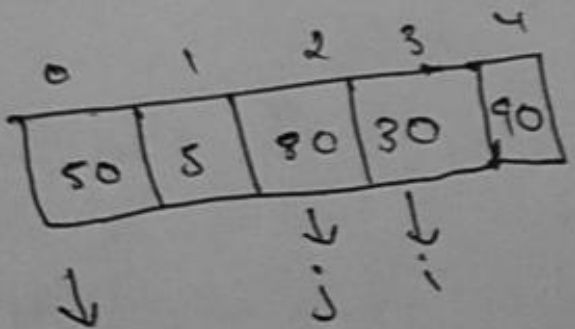
30 is greater than 5, so swap 30 and 5



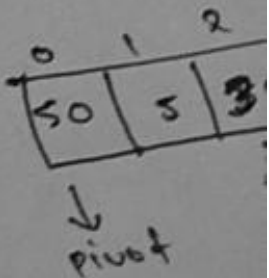
again increment the index i to 2 and decrement

it comes together.

So increment i to 3.



swap (90, 30)



So swap pivot with

ADDITIONAL SHEET

Part-A.

Control abstraction is a procedure where the flow of procedure is clear but the primary procedure has some other procedure which is left is there is undefined.

The control abstraction of divide and conquer is denoted as DAndC(P) where P is the given problem.

Algorithm.

Algorithm DAndC(P)
{
if P is small(P), then return Solution(P);
else
{ Divide P into $P_1, P_2, P_3 \ldots P_k$ $k \geq 1$
Apply DAndC to each subproblems
Return (Combine (DAndC(P_1), DAndC(P_2), DAndC(P_3)...
DAndC(P_k));
}
}

Where small(P) is represent to a small problem.

and form a solution of

→ First we take given problem P and check

is small or not.

→ If the problem is small, then it returns

→ Otherwise it divides problem into subprob

simple subproblem which is further

→ Then recursively apply the DAndC algo

sub problems. Then combine the subproblem

produce a solution to the original prob

In here, functions like solution(), Combine()

not further defined. Only the DAndC() alg

defined.

Control abstraction for greedy strategy.

```
Algorithm Greedy(a, n)
{
    Solution = ϕ ;
    for i = 1 to n do
    {
        s = select(a) ;
        if ( feasible ( solution, s)) then
            Solution = union (solution, s) ;
        else
            reject() ;
    }
    return solution ;
}
```

the sequences of jobs within

* There are set of jobs (J_1, J_2, J_3, J_4) and Deadli
jobs (db_1, db_2, db_3, db_4) and their profits (P_1,
* Each job has single unit of time. So the mi
deadline of each job is 1.
* The jobs are arranged in the order of dea
order of profit.

Algorithm of Job Sequencing.

```
for i = 1 to n do
{
set k = min (Deadline(i), dmax) // where Dead
                                represent to deadline
while k ≥ 1 do
{
if timeslot[k] is empty
{
    timeslot[k] = job(i)
    break;
}
set k = k-1
}
```

Step 1 :- Arrange jobs in descending order of profit.

Step 2 :- Select maximum deadline
Draw a Gantt chart corresponding to it.

Step 4 :- Take each job and place first empty cell before the deadline.

Step 5 :- Fill the Gantt chart and find out the maximum profit of job that presents in the gantt chart.

5) → Minimum Spanning Tree is a spanning tree ~~which~~ which is connected and undirected graph G(V,E) which connects all the vertices together.

→ A single graph may have multiple different spanning tree.

→ The weight of a spanning tree is the sum of weights given to each edge of spanning tree.

→ In minimum ~~tree~~ spanning tree there are two algorithm.

→ Prim's algorithm.

→ Kruskal's algorithm.

executing actions step by step.

Characteristics of algorithm:

→ Input :- It should externally supply zero or one

→ Output :- It results in at least one quantity.

→ Definiteness :- Each instruction should be clear a

→ Finiteness :- An algorithm should terminate at
a finite number of steps.

→ Feasible.

→ Flexibility

→ Efficient.

→ Independent.

7) * Quick sort is a Divide and

* It is an recursive algorithm

* It divides the array into two using an element pivot

* All the elements in the left side is smaller than pivot element.

* All the elements in the right side is greater than the pivot element.

* After dividing into two sub-arrays, the pivot element place at correct position.

* The sub-arrays then perform quick-sort recursively

It performs four steps :-

→ Divide

→ Exchange

→ Conquer

→ Combine.

```
    {
        p = Partition(A, start, end);
        Quicksort(A, start, p-1);
        Quicksort(A, p+1, end);
    }
}

Partition(array a, start, end)
{
    pivot = a[start];
    i = start - 1, j = end - 1;
    while true
    {
        do
        i++ until a[i] ≥ pivot
        j-- until a[i] ≤ pivot
        if (i<j)
            swap(a, i, j);
    }
    swap(a, j, start);
    return j;
}
```